Module 11, Lesson 2

AI 101 for Mechanical Engineers

Loss, gradients and updates

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Structural Outline

1 Loss and Gradients

2 Backpropagation and Optimization

3 Gradient-Descent Calculation

4 Concept Assessment

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1. Loss and Gradients

Defining objective measurement and local change parameters.

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Error Measurement and Parameter Derivatives



Training establishes a quantitative loss objective.

Loss measures deviation from target states.

Gradients describe local parameter-specific changes.

Gradients systematically collect calculated loss derivatives.

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2. Backpropagation and Optimization

Distinguishing derivative computation from parameter updates.

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Backpropagation versus Optimization

Stage	Function	Mechanism
Backpropagation	Computes parameter derivatives	Chain rule through network
Optimizer	Updates system parameters	Gradient descent execution

Backpropagation strictly calculates calculates mathematical derivative values.

Optimizers require gradients to alter weights.

Operations remain mathematically and functionally distinct.

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Formulation of Parameter Updates

$$\theta_{new} = \theta_{old} - \eta \nabla L$$

- θ : System parameter (dimensionless)
- η : Learning rate (dimensionless)
- ∇L : Loss gradient (dimensionless)

Learning rate strictly controls update scaling.

Excessive learning rates induce system overshoot.

Insufficient learning rates decelerate convergence progress.

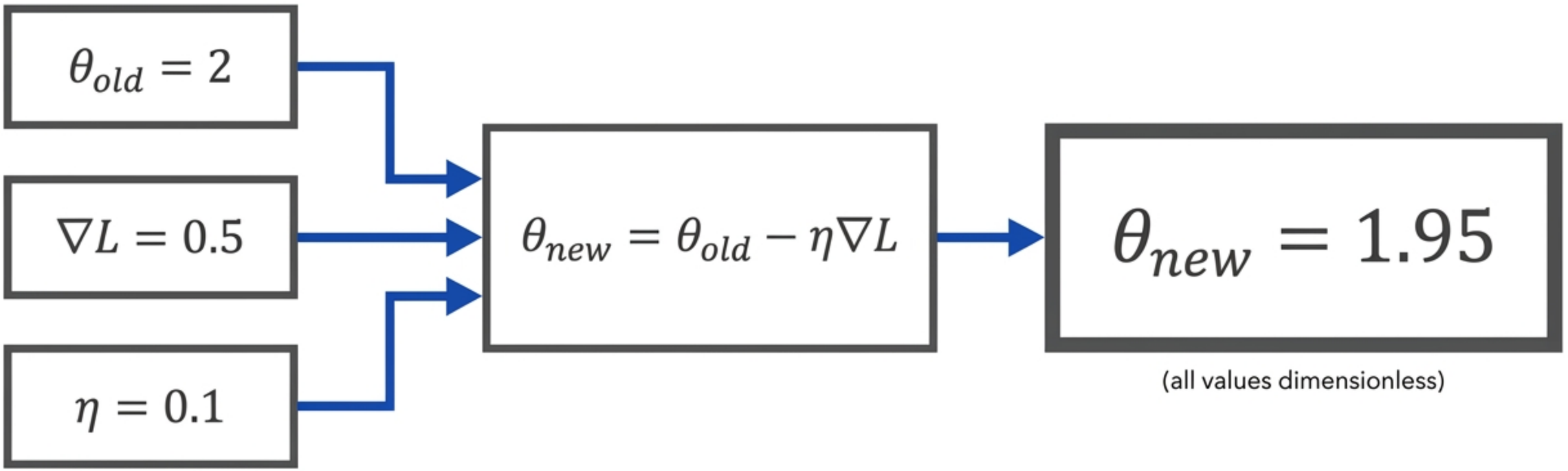
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3. Gradient-Descent Calculation

Executing a single mathematical update step.

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Worked Numerical Example



(all values dimensionless)

Execution demonstrates one isolated mathematical update.

Single updates cannot guarantee global optimization.

Calculations demand strict dimensional unit consistency.

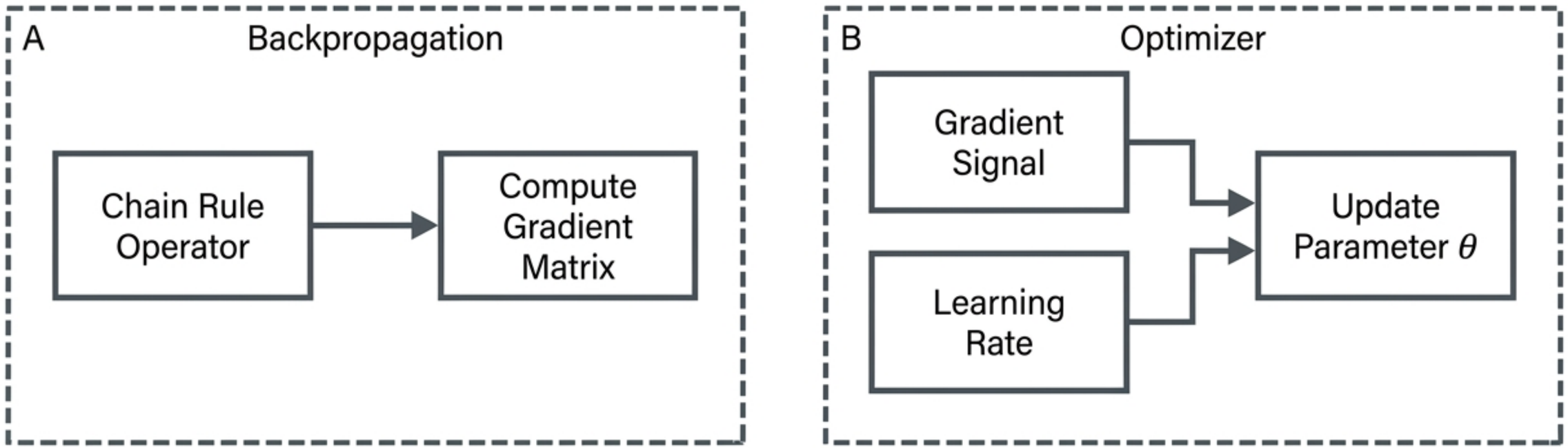
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4. Concept Assessment

Evaluating operational distinctions and numerical outputs.

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Assessment of Core Principles



Backpropagation utilizes chain rules for gradients.

Optimizers subsequently execute required parameter updates.

Mathematical roles must not be conflated.

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Key Points

- Gradients define local change concerning variables.
- Backpropagation calculates derivatives utilizing chain rules.
- Optimizers use gradients to execute updates.
- Gradient descent scales updates via learning rates.

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